

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		40010	
Total Threats Mitigated	35	Total Packets Analyzed		1126839232	
High Priority	11	Malicious Packets		1124727943	
Low Priority	24	True Negatives Passed		34076	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		6	
ML Metrics	Value	SYN Flood		13	
False Positives	0	UDP Flood		6	
FP Rate	0.00%	Uncertain		10	
IF Precision	100.00%				
IF Recall	93.00%	Action		Count	
IF F1-Score	96.38%	Quarantined		10	
IF Accuracy	98.93%	Time Ban		10	
		Blackhole		0	
		Released		3	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.00%	Share of actual attacks successfully flagged
F1-Score	96.38%	Balanced measure combining Precision and Recall
Accuracy	98.93%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	7.00%	Actual attacks that went undetected
True Positive Rate (TPR)	93.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 5623	FN 423
Actual: Normal	FP 0	TN 33653

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	0.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	3279	0	0
Act: ICMP	0	675	0
Act: UDP	0	0	1669

2c. Hybrid Cascaded

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.00%	Share of actual attacks successfully flagged
F1-Score	96.38%	Balanced measure combining Precision and Recall
Accuracy	98.93%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	7.00%	Actual attacks that went undetected
True Positive Rate (TPR)	93.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 5623	FN 423
Actual: Normal	FP 0	TN 33653

2d. Response Latency

Metric	Value	Description
Detection Time	521.3 ms	Average time taken to detect an attack
Mitigation Response Time	16.0 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	3.64%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	11.69%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	16.47%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.19	3	3	2026-09-25 12:06:03	2026-09-25 12:07:23
10.0.0.16	3	2	2026-09-25 12:06:02	2026-09-25 12:07:22
10.0.0.18	3	2	2026-09-25 12:06:02	2026-09-25 12:07:22
10.0.0.17	2	2	2026-09-25 12:06:03	2026-09-25 12:07:13
10.0.0.20	1	1	2026-09-25 12:06:33	2026-09-25 12:06:43
10.0.0.21	1	1	2026-09-25 12:06:33	2026-09-25 12:06:44
10.0.0.22	1	1	2026-09-25 12:06:34	2026-09-25 12:06:44
10.0.0.23	1	1	2026-09-25 12:07:03	2026-09-25 12:07:13

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.24	1	1	2026-09-25 12:07:05	2026-09-25 12:07:15
10.0.0.25	1	1	2026-09-25 12:07:03	2026-09-25 12:07:14

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 12:06:02	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:02	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:03	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:03	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:12	10.0.0.18	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 12:06:12	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 12:06:12	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 12:06:12	10.0.0.18	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 12:06:13	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-25 12:06:13	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Time Ban (1m)
2026-09-25 12:06:13	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-25 12:06:13	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Time Ban
2026-09-25 12:06:33	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:33	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:34	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:43	10.0.0.20	DDoS	UDP Flood	75.6%	High	Time Ban (1m)
2026-09-25 12:06:43	10.0.0.20	DDoS	UDP Flood	75.6%	Critical	Time Ban
2026-09-25 12:06:44	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-25 12:06:44	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban (1m)
2026-09-25 12:06:44	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-25 12:06:44	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban
2026-09-25 12:07:03	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:03	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:05	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:12	10.0.0.18	DDoS	SYN Flood	92.7%	High	Released
2026-09-25 12:07:12	10.0.0.16	DDoS	SYN Flood	92.7%	High	Released
2026-09-25 12:07:13	10.0.0.17	DDoS	SYN Flood	93.3%	High	Time Ban (2m)
2026-09-25 12:07:13	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Released
2026-09-25 12:07:13	10.0.0.23	DDoS	ICMP Flood	90.8%	High	Time Ban (1m)
2026-09-25 12:07:14	10.0.0.23	DDoS	ICMP Flood	90.8%	Critical	Time Ban
2026-09-25 12:07:14	10.0.0.25	DDoS	ICMP Flood	91.2%	High	Time Ban (1m)
2026-09-25 12:07:15	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-25 12:07:15	10.0.0.24	DDoS	ICMP Flood	91.2%	Critical	Time Ban (1m)
2026-09-25 12:07:16	10.0.0.24	DDoS	ICMP Flood	91.2%	High	Time Ban
2026-09-25 12:07:23	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6357	94.0%	Critical	Phase 2	3	10s	2026-09-25 12:07:23	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6384	92.7%	Critical	Phase 1	2	10s	2026-09-25 12:07:22	Attack Stopped
10.0.0.16	SYN Flood	0.6280	92.7%	Critical	Phase 1	2	10s	2026-09-25 12:07:22	Attack Stopped
10.0.0.24	ICMP Flood	0.6451	91.2%	Critical	Phase 2	1	10s	2026-09-25 12:07:15	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6454	91.2%	High	Phase 2	1	11s	2026-09-25 12:07:14	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6071	93.3%	High	Phase 2	2	1m 10s	2026-09-25 12:07:13	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6039	94.0%	Critical	Phase 2	1	1m 10s	2026-09-25 12:07:13	Ban Expired
10.0.0.23	ICMP Flood	0.6433	90.8%	High	Phase 2	1	10s	2026-09-25 12:07:13	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6062	92.7%	High	Phase 2	1	1m 10s	2026-09-25 12:07:12	Ban Expired
10.0.0.16	SYN Flood	0.5952	92.7%	High	Phase 2	1	1m 10s	2026-09-25 12:07:12	Ban Expired
10.0.0.21	UDP Flood	0.6301	77.9%	High	Phase 2	1	11s	2026-09-25 12:06:44	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6250	77.7%	High	Phase 2	1	10s	2026-09-25 12:06:44	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6330	75.6%	High	Phase 2	1	10s	2026-09-25 12:06:43	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6374	90.9%	Critical	Phase 2	1	10s	2026-09-25 12:06:13	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6457	92.5%	Critical	Phase 2	1	10s	2026-09-25 12:06:13	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6464	92.7%	Critical	Phase 2	1	10s	2026-09-25 12:06:12	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6359	92.7%	Critical	Phase 2	1	10s	2026-09-25 12:06:12	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE